

Week 2 - Discovering the Atmosphere 2: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

YOU SHOULD TRY Question 1: Reading atmospheric composition quantitatively

Near the surface, dry air is approximately 78.08% N₂, 20.95% O₂ and 0.93% Ar by mole fraction. Suppose background CO₂ is 420 ppm by mole fraction.

1. Convert 420 ppm to a percentage.
2. Out of 10⁶ air molecules, approximately how many are CO₂ molecules?
3. Why is water vapour normally excluded when quoting the composition of “dry air”?
4. The abundance of CO₂ is far smaller than that of N₂ or O₂. Give one physical reason why abundance alone does not determine atmospheric importance.

Worked answer

Since 1% = 10⁴ ppm,

$$420 \text{ ppm} = 0.0420\%.$$

By definition, 420 molecules in every 10⁶ molecules are CO₂ on average.

Water vapour is excluded from the dry-air reference mixture because its concentration varies strongly with place, height and time, unlike the main permanent gases. A constituent can be physically important even at low abundance if it absorbs radiation strongly in important wavelength bands or participates efficiently in atmospheric chemistry. CO₂ and ozone are examples.

IN CLASS Question 2: An exponential density profile

Assume an isothermal atmosphere with

$$\rho(z) = \rho_0 \exp\left(-\frac{z}{H}\right),$$

where $\rho_0 = 1.20 \text{ kg m}^{-3}$ and $H = 8.0 \text{ km}$.

1. Calculate the density at 10 km.
2. Calculate the density at 50 km.
3. By what factor has the density fallen between the surface and 50 km?
4. State one reason why this simple expression should not be extrapolated uncritically to several hundred kilometres altitude.

Worked answer

At 10 km,

$$\rho = 1.20 \exp(-10/8) = 0.344 \text{ kg m}^{-3}.$$

At 50 km,

$$\rho = 1.20 \exp(-50/8) = 2.32 \times 10^{-3} \text{ kg m}^{-3}.$$

The decrease factor is

$$\frac{1.20}{2.32 \times 10^{-3}} \approx 5.2 \times 10^2,$$

so density is roughly 520 times smaller.

The expression assumes a constant scale height, effectively a simplified isothermal atmosphere with nearly constant gravity and composition. Temperature and composition change strongly with altitude, so one constant H cannot describe the whole upper atmosphere.

EXTRA CHALLENGE Question 3: How high before density is only ten per cent?

Using the same exponential model as in Question 2, calculate the altitude at which the density is 10% of its sea-level value. Then calculate the altitude at which it is 1% of its sea-level value.

Worked answer

For $\rho/\rho_0 = 0.10$,

$$0.10 = e^{-z/H}, \quad z = H \ln 10 = 8.0 \times 2.303 = 18.4 \text{ km}.$$

For $\rho/\rho_0 = 0.01$,

$$z = H \ln 100 = 8.0 \times 4.605 = 36.8 \text{ km}.$$

Thus the idealised density reaches about 10% at 18.4 km and 1% at 36.8 km.

IN CLASS Question 4: Diagnose the atmospheric layers from a temperature profile

The table gives a deliberately simplified mean temperature profile.

Altitude (km)	0	10	20	50	85
Temperature (°C)	15	-55	-55	0	-90

1. Calculate the mean lapse rate between 0 and 10 km.
2. Identify the broad atmospheric layer represented by 0–10 km.
3. Which layer is represented by the temperature increase from 20 to 50 km?
4. What feature near 10–20 km separates these two layers?
5. Why can ozone absorption produce a temperature increase with height in the stratosphere?

Worked answer

Temperature falls by 70 K over 10 km, giving a mean lapse rate

$$\Gamma = \frac{70}{10} = 7.0 \text{ K km}^{-1}.$$

The 0–10 km region is the troposphere. The warming from about 20 to 50 km is characteristic of the stratosphere. The transition region is the tropopause/lower stratosphere, where the tropospheric decrease of temperature with height becomes very small or reverses.

Ozone absorbs solar ultraviolet radiation. The absorbed energy heats the surrounding air, providing a radiative heat source within the stratosphere and allowing temperature to increase with height.

YOU SHOULD TRY Question 5: Same molecule, different altitude: ozone

Two instruments report enhanced ozone. Site A is a surface monitoring station in a polluted summer air mass. Site B is a balloon instrument at 25 km altitude.

1. At which site is enhanced ozone primarily a health/air-quality concern?
2. At which site does ozone play its major protective role against solar ultraviolet radiation?
3. State the physical reason for each answer in one or two sentences.

Worked answer

Site A is the air-quality concern. Near the surface, ozone is a strong oxidant and elevated concentrations can damage respiratory health, vegetation and materials.

Site B represents the protective stratospheric role. Ozone at stratospheric altitudes absorbs biologically damaging ultraviolet radiation before it reaches the surface. The molecule is the same; its consequence depends strongly on where in the atmosphere it is located.

YOU SHOULD TRY Question 6: Why does the Antarctic ozone hole develop in spring?

The Antarctic ozone hole results from a sequence of linked physical and chemical conditions:

CFC-derived chlorine in the stratosphere → cold, isolated winter vortex and PSCs → chlorine is chemically primed on PSC surfaces → spring sunlight activates chlorine → rapid catalytic ozone loss.

Use this process chain to answer the following questions.

1. Explain in one sentence why the polar vortex is important.
2. Explain in one sentence why PSCs are important.
3. Why does the most rapid ozone loss occur after sunlight returns in spring rather than during the dark Antarctic winter?
4. What eventually limits the severe seasonal ozone loss as spring progresses?

Worked answer

The polar vortex isolates Antarctic stratospheric air from lower latitudes and helps maintain the very low temperatures required for PSC formation.

PSCs provide particle surfaces on which heterogeneous reactions convert relatively inactive chlorine reservoir compounds into forms that are readily photochemically activated.

During the dark polar winter, the necessary sunlight is absent. When sunlight returns in spring, photochemistry releases reactive chlorine radicals, allowing rapid catalytic ozone destruction.

As spring progresses, the polar vortex weakens and breaks down, the stratosphere warms, PSCs disappear, and mixing with lower-latitude ozone-rich air increases. These changes reduce the conditions that sustain the most severe ozone depletion.

Modelling connection.

The following two short problems preview parcel and cloud physics that will be developed in more detail later in EART23001. Students who also take EART22001 will meet related ideas in a parcel-model practical, but no knowledge of that unit is assumed here.

IN CLASS Question 7: A rising parcel: pressure, cooling and saturation (preview)

A dry air parcel starts at $p_1 = 1000$ hPa and $T_1 = 290$ K and rises to a level where $p_2 = 850$ hPa. Treat the ascent as dry adiabatic and use

$$T_2 = T_1 \left(\frac{p_2}{p_1} \right)^{R_d/c_p}, \quad \frac{R_d}{c_p} = 0.286.$$

At the initial level the water-vapour partial pressure is $e = 12.0$ hPa. For this preview, assume e changes little before saturation is reached. Use the supplied values $e_s(290 \text{ K}) = 19.2$ hPa and $e_s(T_2) = 11.7$ hPa.

1. Calculate T_2 .
2. Calculate the initial relative humidity, $RH = 100e/e_s$.
3. Using the supplied $e_s(T_2)$, calculate the saturation ratio $\mathcal{S} = e/e_s$ at the upper level and the supersaturation $s = (\mathcal{S} - 1) \times 100\%$.
4. Explain in one or two sentences why rising motion can lead to cloud formation.

Worked answer

The dry-adiabatic temperature is

$$T_2 = 290 \left(\frac{850}{1000} \right)^{0.286} = 276.8 \text{ K}.$$

Initially,

$$RH = 100 \frac{12.0}{19.2} = 62.5\%.$$

At the upper level,

$$\mathcal{S} = \frac{12.0}{11.7} = 1.0256, \quad s = 2.56\%.$$

The numerical supersaturation is deliberately exaggerated by the simplifying assumption that e does not change before condensation; a real parcel model updates both vapour content and temperature continuously and condensation limits supersaturation. The physical chain is the important point: ascent lowers pressure, the parcel expands and cools adiabatically, $e_s(T)$ falls rapidly, and saturation can be reached.

YOU SHOULD TRY Question 8: Same liquid water, different droplet number (preview)

A model cloud contains the same liquid-water content in two simulations. Simulation B has eight times the droplet number concentration of Simulation A. Assume identical spherical droplets within each simulation.

1. Using “number $\times$ average droplet mass”, explain why constant liquid-water content implies $N_d r^3 = \text{constant}$.
2. By what factor does mean radius r change when N_d increases by a factor of 8?
3. Total projected droplet area is proportional to “number $\times$ average projected area”, $N_d r^2$. By what factor does the total projected area change?
4. Show that, at fixed liquid-water content, total projected droplet area is proportional to $N_d^{1/3}$. Why is this useful for thinking about the interaction of a cloud with solar radiation?

Worked answer

Droplet mass is proportional to r^3 , so total liquid-water mass per unit volume scales as $N_d r^3$. Keeping it fixed gives

$$N_d r^3 = \text{constant}, \quad r \propto N_d^{-1/3}.$$

Increasing N_d by 8 therefore reduces r by $8^{-1/3} = 1/2$.

The total projected area changes by

$$\frac{(N_d r^2)_B}{(N_d r^2)_A} = 8 \left(\frac{1}{2}\right)^2 = 2.$$

Thus the same liquid water split among eight times as many droplets gives droplets half the radius but twice the total projected area. More generally,

$$N_d r^2 \propto N_d N_d^{-2/3} = N_d^{1/3}.$$

For visible wavelengths, greater total projected droplet area generally gives greater extinction of solar radiation through the cloud. This provides a simple physical link between droplet number and cloud radiative properties; the fuller cloud-radiation treatment comes later in the course.